

Mathematical Summary of pyMacroIO: Dynamic Disequilibrium Input-Output Model

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This document gives an overview of the simple single-region Dynamic Disequilibrium Input-Output (IO) model. This follows descriptions in [1,2,3] closely, but the implementation here is organised around a benchmark-consistent transition map: in the absence of shocks, the calibrated base year is an exact fixed point of the one-period dynamics. The model is discrete-time and sector-level. It includes Leontief (traditional or adapted), linear, or CES production; inventories; labour hiring and firing; a household demand block with persistent consumption; and two exogenous shocks: the consumption shock and the input-availability shock. Base-year example data are taken from a single regional data set. The Python implementation is provided in the `pyMacroIO` module; the default data file is `data/example_data.pkl` [4].

This summary reflects the current repository state.

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1 Model Structure

1.1 Basic Setup

The model is defined over N sectors and T time periods within a single region only. Gross output at time t is denoted by $\mathbf{x}_t \in \mathbb{R}^N$, the technical coefficient matrix by $\mathbf{A} \in \mathbb{R}^{N \times N}$, total demand by $\mathbf{d}_t \in \mathbb{R}^N$, and final demand by $\mathbf{f}_t \in \mathbb{R}^N$. No regional index is used, since the system is strictly single-region.

1.2 Economic Balance Constraint

The fundamental static balance is

$$\mathbf{x} = \mathbf{A}\mathbf{x} + \mathbf{f}, \quad (1)$$

from which

$$\mathbf{x} = (\mathbf{I} - \mathbf{A})^{-1}\mathbf{f} = \mathbf{L}\mathbf{f} \quad (2)$$

is obtained, where $\mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1}$ is the Leontief inverse. In the dynamic model, this structure underlies calibration and the identification of essential inputs; within-period equilibrium is enforced by the sequential determination of labour, demand, production, intermediate consumption, and inventory update.

2 Data and Parameter Calibration

2.1 Technical Coefficient Matrix

The technical coefficient matrix $\mathbf{A}$ is constructed from base-year intermediate flows $\mathbf{Z}_0$ and base-year gross output $\mathbf{x}_0$. The gross output $\mathbf{x}_0$ is obtained from the row identity: the output of each sector equals the sum of its intermediate sales plus total final demand. Then

$$A_{ij} = \frac{Z_{0,ij}}{x_{j,0}}, \quad (3)$$

with the convention that each column sum of intermediate inputs is less than output:

$$\sum_{i=1}^N A_{ij} < 1 \quad \forall j. \quad (4)$$

2.2 Essential Inputs Identification

For the adapted Leontief production function, a binary importance matrix $\mathbf{A}_{\text{essential}}$ is built. The primary method is the combined Linkage Method; a value-based fallback is used when the Leontief inverse cannot be computed [5].

2.2.1 Combined Linkage Method (Primary)

With $\mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1}$ and $\mathbf{G} = (\mathbf{I} - \mathbf{A}^T)^{-1}$:

$$\text{Backward Linkage}_j = \sum_{i=1}^N L_{ij}, \quad (5)$$

$$\text{Forward Linkage}_i = \sum_{j=1}^N G_{ij}. \quad (6)$$

The linkages are normalised by their means (with a small constant $\varepsilon > 0$ to avoid division by zero):

$$\text{Norm Backward}_j = \frac{\text{Backward Linkage}_j}{\text{Backward Linkage} + \varepsilon}, \quad (7)$$

$$\text{Norm Forward}_i = \frac{\text{Forward Linkage}_i}{\text{Forward Linkage} + \varepsilon}. \quad (8)$$

The combined linkage is $\text{Combined Linkage}_{ij} = \text{Norm Forward}_i \times \text{Norm Backward}_j$. The binary indicator is

$$A_{\text{essential},ij} = \begin{cases} 1 & \text{if Combined Linkage}_{ij} > \theta \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

where θ is a (threshold) parameter.

2.2.2 Value-Based Fallback

If matrix inversion fails, essential inputs are identified by value share:

$$A_{\text{essential},ij} = \begin{cases} 1 & \text{if } \frac{A_{ij}}{\sum_{k=1}^N A_{kj} + \varepsilon} \geq \theta_{\text{value}} \\ 0 & \text{otherwise} \end{cases} \quad (10)$$

where θ_{value} is a value-share threshold.

2.3 Value Added Decomposition

Value added in the base year is decomposed as

$$\text{VA}_j = L_j + \text{cap}_j + \text{tax}_j + \text{other}_j + \Pi_j, \quad (11)$$

where L_j is labour cost, cap_j capital cost, tax_j taxes, other_j other costs, and Π_j profits. Value added is taken from the column identity (output minus domestic intermediate inputs minus imports). Base-year labour $\ell_{j,0}$, capital shares λ_j^{cap} , tax shares λ_j^{tax} , and import shares λ_j^{imp} are used, so that profits are computed as

$$\Pi_{j,t} = x_{j,t} - \sum_{i=1}^N Z_{ij,t} - \ell_{j,t} - (\lambda_j^{\text{cap}} + \lambda_j^{\text{tax}} + \lambda_j^{\text{imp}}) x_{j,t}. \quad (12)$$

3 Dynamic Model Equations

3.1 Time Step and Consumption Persistence

Two time frequencies are considered: daily and quarterly. The persistence parameter ρ_1 (and $\rho_0 = 1 - \rho_1$) in the consumption function is set from the chosen frequency. For daily frequency, $\rho_1 = 1 - (1 - \bar{\rho}) dt$ and $\rho_0 = 1 - \rho_1$ for a baseline $\bar{\rho}$ with $dt = 1/90$. For quarterly frequency, the implementation uses the fixed calibration $(\rho_0, \rho_1) = (0.4, 0.6)$ with $dt = 0.25$. The parameter dt is used only for this calibration and does not appear in the dynamic equations.

3.2 Labour Adjustment Mechanism

Labour is adjusted gradually towards the output level implied by the previous period's non-labour constraints, using sector-specific hiring and firing speeds $\gamma_{\text{hire},j}$ and $\gamma_{\text{fire},j}$. When hiring and firing are switched off, $\ell_{j,t} = \ell_{j,t-1}$ is imposed. Otherwise the following quantities are used:

$$\ell_{\text{cap},j,t} = \ell_{j,0} \cdot \max(\min(1 - \delta_{j,t}, 1), \underline{\delta}), \quad (13)$$

$$\ell_{\text{cap},\text{max},j} = \bar{\varsigma} \ell_{j,0}, \quad \ell_{\text{cap},\text{min},j} = \underline{\varsigma} \ell_{j,0}, \quad (14)$$

$$\ell_{\text{cap},j,t}^{\text{new}} = \begin{cases} \min(\ell_{\text{cap},\text{max},j}, \max(\ell_{\text{cap},j,t}, \ell_{j,t-1})), & \delta_{j,t} > 0, \\ \ell_{\text{cap},\text{max},j}, & \delta_{j,t} = 0, \end{cases} \quad (15)$$

$$\text{share}_j = \frac{\ell_{j,0}}{x_{j,0}}, \quad (16)$$

$$\hat{x}_{j,t} = \min(\text{input_constraint}_{j,t-1}, \text{demand}_{j,t-1}, \text{output_constraint}_{j,t-1}), \quad (17)$$

$$\hat{\ell}_{j,t} = \text{clip}(\text{share}_j \hat{x}_{j,t}, \ell_{\text{cap},\text{min},j}, \ell_{\text{cap},j,t}^{\text{new}}), \quad (18)$$

$$\Delta_j = \hat{\ell}_{j,t} - \ell_{j,t-1}. \quad (19)$$

The adjustment uses $\gamma_{\text{hire},j}$ when $\Delta_j > 0$ and $\phi \gamma_{\text{fire},j}$ when $\Delta_j \leq 0$:

$$\ell_{j,t} = \text{clip}(\ell_{j,t-1} + \gamma_j \cdot \Delta_j, \ell_{\text{cap},\text{min},j}, \ell_{\text{cap},j,t}^{\text{new}}), \quad (20)$$

where $\gamma_j = \gamma_{\text{hire},j}$ if $\Delta_j > 0$ and $\gamma_j = \phi \gamma_{\text{fire},j}$ otherwise, and $\phi \in (0, 1]$ is a firing-speed damping factor. The bounds on δ and the capacity scale factors $\underline{\varsigma}$, $\bar{\varsigma}$ are model parameters.

3.3 Household Consumption and Savings

Household disposable income is the sum of adjusted labour income and profit income:

$$Y_t^h = \beta L_0 + (1 - \beta) L_t + \Pi_t, \quad (21)$$

where $L_t = \sum_j \ell_{j,t}$, L_0 is base-year labour income, $\Pi_t = \sum_j \Pi_{j,t}$, and $\beta \in [0, 1]$ is the benefits parameter. In the baseline calibration, observed household consumption from the data is preserved; when the savings rate is not supplied explicitly, it is inferred from base-year household income and spending. Household savings are treated as an ex post accounting outcome. Two closure modes are implemented.

Define the closure-dependent income signal as

$$Y_{t-1}^{h,\text{sig}} = \begin{cases} \xi_t Y_0^h, & \text{if household closure mode} = \text{return_to_base}, \\ Y_{t-1}^h, & \text{if household closure mode} = \text{scarred}. \end{cases} \quad (22)$$

Then current and expected household consumption capacities are

$$\bar{C}_t^{\text{cur}} = (1 - \nu)(1 - s) \max(Y_{t-1}^{h,\text{sig}}, \varepsilon), \quad (23)$$

$$\bar{C}_t^{\text{exp}} = (1 - \nu)(1 - s) \max(\xi_t Y_0^h, \varepsilon), \quad (24)$$

where $s \in [0, 1)$ is the household savings rate, $\nu \in [0, 1)$ is the extra-expenditure coefficient, and ξ_t is the expectations term. Aggregate desired consumption is updated in benchmark-relative form:

$$C_t = C_0 \exp \left(\rho_1 \log \frac{C_{t-1}^{\text{sig}}}{C_0} + \frac{\rho_0}{2} \log \frac{\bar{C}_t^{\text{cur}}}{C_0} + \frac{\rho_0}{2} \log \frac{\bar{C}_t^{\text{exp}}}{C_0} \right), \quad (25)$$

with a lower floor relative to baseline consumption and an upper cap at $\bar{C}_t^{\text{cur}}$. The log arguments are floored by fractions of baseline consumption (model parameters). This formulation makes the benchmark an exact fixed point when $Y_t^h = Y_0^h$, $\xi_t = 1$, and shocks are absent. Here

$$\rho_0 + \rho_1 = 1, \quad (26)$$

$$\text{MPC} = \frac{\sum_{j=1}^N c_{j,0}}{Y_0^h}. \quad (27)$$

The lagged-consumption signal is also closure-dependent:

$$C_{t-1}^{\text{sig}} = \begin{cases} C_0, & \text{if household closure mode} = \text{return_to_base}, \\ C_{t-1}, & \text{if household closure mode} = \text{scarred}. \end{cases} \quad (28)$$

Hence the **return_to_base** closure anchors the beginning-of-period income signal to benchmark income and neutralises the lagged-consumption term, so temporary shocks do not create permanent endogenous demand scars. The **scarred** closure instead uses lagged realised income and lagged realised aggregate consumption, so temporary shocks may leave persistent demand scars. Comparing these closures is therefore a structural sensitivity exercise rather than ordinary parameter uncertainty.

Sectoral consumption demand is

$$c_{j,t}^d = \theta_{j,t} \cdot C_t \cdot (1 - \epsilon_t), \quad (29)$$

with $\theta_{j,t} = c_{j,0} / \sum_k c_{k,0}$ fixed from base-year shares.

3.4 Inventory Management and Orders

Target inventories are proportional to base-year intermediate flows and a sector-specific coverage n_j (in days):

$$S_{\text{target},ij} = Z_{0,ij} \cdot n_j = A_{ij} x_{j,0} n_j. \quad (30)$$

Orders for intermediate inputs are

$$O_{ij,t} = A_{ij} d_{j,t-1} + \frac{\max\{S_{\text{target},ij} - S_{ij,t-1}, 0\}}{\tau_j}, \quad O_{ij,t} = \max(0, O_{ij,t}), \quad (31)$$

where τ_j is the inventory-adjustment speed of the receiving sector j . Each τ_j is sector-specific and lies in an interval $[\underline{\tau}, \bar{\tau}]$ prescribed by the model. In particular, inventory correction is one-sided: shortages are rebuilt gradually, while excess stocks are worked off by current use rather than by forcing orders below use demand.

3.5 Production Functions

Capacity is determined by labour: $x_{\text{cap},j,t} = (\ell_{j,t}/\ell_{j,0})x_{j,0}$, and may be further limited by an exogenous output constraint $x_{\text{cap},j,t}^{\text{out}}$. The effective capacity is $\min(x_{\text{cap},j,t}, x_{\text{cap},j,t}^{\text{out}})$. Production is given by the minimum of capacity, input-determined output, and demand. The model admits four production types: traditional Leontief, adapted Leontief (with essential inputs), linear, and CES.

3.5.1 Traditional Leontief

$$x_{j,t} = \min \left(\text{capacity}_{j,t}, \min_{i:A_{ij}>0} \frac{S_{ij,t-1}}{A_{ij}}, d_{j,t} \right). \quad (32)$$

3.5.2 Adapted Leontief with Essential Inputs

Essential inputs are those with $A_{\text{essential},ij} > \theta_{\text{ess}}$, for a threshold θ_{ess} . They remain hard bottlenecks, while the non-essential inputs are pooled more flexibly:

$$x_{\text{input},j}^{\text{ess}} = \min_{i:A_{\text{essential},ij}>\theta_{\text{ess}}, A_{ij}>0} \frac{S_{ij,t-1}}{A_{ij}}, \quad x_{\text{input},j}^{\text{noness}} = \frac{\sum_{i \in \mathcal{N}_j} A_{ij} (S_{ij,t-1}/A_{ij})}{\sum_{i \in \mathcal{N}_j} A_{ij}}, \quad (33)$$

where $\mathcal{N}_j = \{i : A_{\text{essential},ij} \leq \theta_{\text{ess}}, A_{ij} > 0\}$. The input constraint is $x_{\text{input},j} = \min(x_{\text{input},j}^{\text{ess}}, x_{\text{input},j}^{\text{noness}})$, with the usual convention that an empty set gives $+\infty$. Then

$$x_{j,t} = \min(\text{capacity}_{j,t}, x_{\text{input},j}, d_{j,t}). \quad (34)$$

With the default (combined-linkage) method, $\mathbf{A}_{\text{essential}}$ is binary $\{0, 1\}$ and therefore cleanly separates essential from non-essential inputs.

3.5.3 Linear Production

$$x_{j,t} = \min \left(\text{capacity}_{j,t}, \frac{\sum_{i=1}^N S_{ij,t-1}}{\sum_{i=1}^N A_{ij}}, d_{j,t} \right). \quad (35)$$

3.5.4 CES Production

For the CES specification, all inputs with $A_{ij} > 0$ enter a genuine substitution aggregator built from their implied output capacities $q_{ij,t} = S_{ij,t-1}/A_{ij}$ and technical-coefficient weights $\omega_{ij} = A_{ij}/\sum_{k:A_{kj}>0} A_{kj}$. For elasticity $\sigma > 0$, define $\rho = (\sigma - 1)/\sigma$. The CES input constraint is

$$x_{\text{input},j}^{\text{CES}} = \begin{cases} \left(\sum_{i:A_{ij}>0} \omega_{ij} q_{ij,t}^\rho \right)^{1/\rho}, & \sigma \neq 1, \\ \exp \left(\sum_{i:A_{ij}>0} \omega_{ij} \log q_{ij,t} \right), & \sigma = 1. \end{cases} \quad (36)$$

Output is then

$$x_{j,t} = \min(\text{capacity}_{j,t}, x_{\text{input},j}^{\text{CES}}, d_{j,t}). \quad (37)$$

Unlike adapted Leontief, the CES branch does not use $\mathbf{A}_{\text{essential}}$; its difference comes from substitution elasticity rather than a binary bottleneck partition.

3.6 Rationing

Absent shocks, industries can always meet total demand, i.e. $x_{j,t} = d_{j,t}$. However, in the presence of production capacity and/or input bottlenecks, industries' output may be smaller than total demand (i.e., $x_{j,t} < d_{j,t}$), in which case output is rationed across customers. Simple proportional rationing is assumed. The final delivery from industry i to industry j is the share of orders received,

$$Z_{ij,t} = O_{ij,t} \frac{x_{i,t}}{d_{i,t}}. \quad (38)$$

A share of household demand is received,

$$c_{i,t} = c_{i,t}^d \frac{x_{i,t}}{d_{i,t}}, \quad (39)$$

and the realised final consumption of agents with exogenous final demand is

$$f_{i,t} = f_{i,t}^d \frac{x_{i,t}}{d_{i,t}}. \quad (40)$$

(When the model disaggregates final demand into government, investment, exports, etc., the same sector-level share $x_{i,t}/d_{i,t}$ applies to each component.)

3.7 Intermediate Consumption

Desired intermediate use is given by orders scaled by a supply/demand factor. Under proportional rationing, when $x_{i,t} < d_{i,t}$, the supply/demand share $x_{i,t}/d_{i,t}$ for the supplying sector i yields the rationing formulae above. Two allocation rules are implemented:

- **Proportional across total demand:** $s_i = x_{i,t}/d_{i,t}$ for supplier i ; then $Z_{ij,t} = O_{ij,t} s_i$.
- **Firm-priority rule:** Per supplier i , $s_i = \min(1, x_{i,t}/\sum_{k=1}^N O_{ik,t})$; then $Z_{ij,t} = O_{ij,t} s_i$.

In both cases, factors are clipped to $[0, 1]$. Beginning-of-period inventories constrain production through the production function, but they do not cap new receipts.

3.8 Final Consumption (Realised)

Under the by-recipient rule, realised consumption is $s_j = x_{j,t}/d_{j,t}$ per sector j , so $c_{j,t} = c_{j,t}^d \cdot s_j$. Under the by-supplier rule, a residual supply/demand ratio is applied: for each sector i ,

$$s_i = \frac{x_{i,t} - \sum_{k=1}^N Z_{ik,t}}{d_{i,t} - \sum_{k=1}^N Z_{ik,t}}, \quad \text{clipped to } [0, 1], \quad c_{i,t} = c_{i,t}^d \cdot s_i, \quad (41)$$

i.e. output net of intermediate deliveries over demand net of those deliveries.

3.9 Inventory Update

The inventory of input i held by sector j is updated according to

$$S_{ij,t} = \max(0, S_{ij,t-1} + Z_{ij,t} - A_{ij} x_{j,t}). \quad (42)$$

In a Leontief production function, where every input is critical, the maximum operator would be redundant since production could never continue once inventories are run down. It is necessary for other production functions since industries can produce even after inventories of one or more non-critical inputs are depleted; thus the maximum operator avoids inventories becoming negative.

3.10 Profit and GDP

Profits are computed using labour and value-added shares (capital, tax, import) from the base year:

$$\Pi_{j,t} = x_{j,t} - \sum_{i=1}^N Z_{ij,t} - \ell_{j,t} - (\lambda_j^{\text{cap}} + \lambda_j^{\text{tax}} + \lambda_j^{\text{imp}}) x_{j,t}. \quad (43)$$

GDP (value added, output minus intermediate consumption, with import adjustment) is

$$\text{GDP}_t = \sum_{j=1}^N \left(x_{j,t} - \sum_{i=1}^N Z_{ij,t} - \lambda_j^{\text{imp}} x_{j,t} \right). \quad (44)$$

3.11 Savings

Aggregate savings are computed as

$$S_t = Y_t^h - \sum_{j=1}^N c_{j,t} - E_t, \quad (45)$$

where

$$E_t = \frac{\nu}{1-\nu} \sum_{j=1}^N c_{j,t} \quad (46)$$

and $\nu \in [0, 1)$ is the extra-expenditure coefficient. In both implemented closures this is reported as an accounting output; it does not feed back into future demand through a wealth state.

3.12 Final Demand and Aggregate Demand

Total demand for sector j in period t is

$$d_{j,t} = c_{j,t}^d + \sum_{k=1}^N O_{jk,t} + f_{\text{gov},j,t} + f_{\text{inv},j,t} + f_{\text{invnt},j,t} + f_{\text{exp},j,t} + f_{\text{other},j,t}, \quad (47)$$

where $c_{j,t}^d$ is consumption demand from the household budget block, $\sum_k O_{jk,t}$ is intermediate demand for sector j 's output (orders placed with sector j by all sectors k), and f_{gov} , f_{inv} , f_{invnt} , f_{exp} , f_{other} are government, investment, inventory (final demand), exports, and any other final demand. In the single-region model one such vector per period is used.

4 Consumption Shock (Simple Example)

In this simple example, the consumption shock is active in the example run. For $t \in [t_{\text{start}}, t_{\text{start}} + \text{duration})$, the shock intensity $\epsilon_t \in [0, 1)$ is set (e.g. $\epsilon_t = 0.2$ for a 20% cut). Sectoral consumption demand is scaled down via

$$c_{j,t}^d = \theta_{j,t} C_t (1 - \epsilon_t). \quad (48)$$

The default example uses intensity 0.2, duration 3 periods, and start period 2. When demand is the binding constraint, all production functions yield identical results.

5 Input-Availability Shock

An input-availability shock is modelled as a supplier-side output-capacity reduction for a chosen input sector. For each period t in the shock window, supplier i^* faces the temporary cap

$$x_{i^*,t}^{\text{out}} \leq (1 - \zeta) x_{i^*,0}, \quad (49)$$

where $\zeta \in [0, 1)$ is the reduction fraction. The supplier's reduced output is then rationed across intermediate and final demand in the usual way, so downstream sectors feel the shock through lower deliveries and subsequent inventory drawdown rather than through direct deletion of their inventories. The implementation distinguishes between a moderate example case and a tighter stress case. The example helper uses a key supplier chosen by largest row sum of $\mathbf{A}$, reduction 0.3, duration 3 periods, start period 2, and 5 days of inventory cover. The production-function comparison helper uses a stress specification with reduction 0.5 and 1 day of inventory cover, because structural differences between Leontief, adapted Leontief, CES, and linear production are otherwise often hidden by generous inventories.

6 Monte Carlo Uncertainty Analysis

Parameters may be drawn from specified distributions; for each draw the model is solved and GDP, aggregate consumption, and gross output by sector and time are stored.

6.1 Parameter Distributions

Typical choices are uniform distributions over prescribed intervals for:

1. **Consumption persistence** ρ_1 (with $\rho_0 = 1 - \rho_1$),
2. **Labour hiring speed** $\gamma_{\text{hire},j}$ per sector,
3. **Labour firing speed** $\gamma_{\text{fire},j}$ per sector,
4. **Inventory adjustment** τ_j per sector,
5. **Savings rate** s (used in the household budget rule; if omitted, it is inferred from observed base-year household spending and income so the input data remain the baseline calibration).

6.2 Uncertainty Metrics

For each stored variable Y and time t , the mean, standard deviation, and selected quantiles are computed over valid simulations (invalid or failed draws are excluded):

$$\mu_{Y,t} = \frac{1}{M} \sum_{i=1}^M Y_{i,t}, \quad \sigma_{Y,t} = \sqrt{\frac{1}{M} \sum_{i=1}^M (Y_{i,t} - \mu_{Y,t})^2}, \quad (50)$$

$$q_{Y,t}^{(p)} = \text{percentile}(Y_{1,t}, \dots, Y_{M,t}, p). \quad (51)$$

Selected quantiles are used for reporting and visualisation.

6.3 Interpretation of the Bands

The reported quantiles and bands are intended to reflect parameter uncertainty (uncertainty in the structural parameters listed above). Uncertainty about the shock itself (timing, size, or duration) is not included. Because the same parameters govern the dynamics in every period, the spread across simulations may persist or widen after the shock has ended; the bands do not necessarily narrow once the shock is removed.

7 Model Assumptions and Consistency

7.1 Economic Balance

The row identity of the base-year accounts ($\mathbf{x}_0 = \mathbf{Z}_0 \mathbf{1} + \mathbf{f}_0$) and the column identity (output equals domestic intermediate inputs plus imports plus value added) are assumed to hold for calibration. The dynamic simulation then evolves by sequential rationing and accounting updates built from those benchmark objects; it should not be read as re-imposing the full static IO system as an equilibrium condition in every period.

7.2 Parameter Bounds

Sector parameters are required to satisfy

$$\underline{\gamma}_{\text{hire}} \leq \gamma_{\text{hire},j} \leq \bar{\gamma}_{\text{hire}}, \quad \underline{\gamma}_{\text{fire}} \leq \gamma_{\text{fire},j} \leq \bar{\gamma}_{\text{fire}}, \quad \underline{\tau} \leq \tau_j \leq \bar{\tau} \quad \forall j, \quad (52)$$

for model-given bounds $\underline{\gamma}_{\text{hire}}, \bar{\gamma}_{\text{hire}}, \underline{\gamma}_{\text{fire}}, \bar{\gamma}_{\text{fire}}, \underline{\tau}, \bar{\tau}$. Consumption parameters satisfy $\rho_0 + \rho_1 = 1$. The benefits parameter β is in $[0, 1]$.

8 Order of Operations (Single Period)

The model is solved as a discrete-time sequential simulation. For each period $t = 1, \dots, T-1$, the order of operations is:

1. **Labour:** $\ell_{j,t}$ is updated from $\ell_{j,t-1}$, $x_{j,t-1}$, $\delta_{j,t}$, and the previous period's capacity, input, and demand constraints, using the hiring/firing rule above.
2. **Capacity:** $x_{\text{cap},j,t} = (\ell_{j,t}/\ell_{j,0}) x_{j,0}$; output constraints are applied if present.

3. **Consumption demand:** C_t and $c_{j,t}^d$ are computed from the closure-dependent household income signal $Y_{t-1}^{h,\text{sig}}$, the closure-dependent lagged consumption signal C_{t-1}^{sig} , expectations ξ_t , the savings rule, and the consumption shock ϵ_t .
4. **Orders:** $O_{ij,t}$ is computed from $\mathbf{A}$, $d_{\cdot,t-1}$, τ , S_{target} , and $S_{\cdot,t-1}$.
5. **Aggregate demand:** $d_{j,t} = c_{j,t}^d + \sum_k O_{jk,t} + f_{\text{gov},j,t} + f_{\text{inv},j,t} + f_{\text{invnt},j,t} + f_{\text{exp},j,t} + f_{\text{other},j,t}$.
6. **Production:** $x_{j,t}$ is determined from capacity, input constraints, and demand, according to the chosen production type.
7. **Intermediate consumption:** $Z_{ij,t}$ is determined from $O_{\cdot,t}$, $d_{\cdot,t}$, $x_{\cdot,t}$, and the supplier-side allocation rule. When rationing shocks are active for period t , deliveries from rationed suppliers are scaled proportionally.
8. **Realised consumption:** $c_{j,t}$ is determined from $c_{\cdot,t}^d$, $d_{\cdot,t}$, $x_{\cdot,t}$, $Z_{\cdot,t}$, and the allocation rule. When rationing shocks are active for period t , consumption from rationed suppliers is scaled proportionally.
9. **Inventory:** $S_{ij,t} = \max(0, S_{ij,t-1} + Z_{ij,t} - A_{ij} x_{j,t})$.
10. **Accounting:** $\Pi_{j,t}$, aggregate household savings S_t , and GDP_t are computed from the formulae above.

9 References

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